



Real-time feasible multi-objective optimization based nonlinear model predictive control of particle size and shape in a batch crystallization process



Yankai Cao, David Acevedo, Zoltan K. Nagy, Carl D. Laird *

School of Chemical Engineering, Purdue University, 480 Stadium Mall Drive, West Lafayette, IN 47907, USA

ABSTRACT

This paper presents nonlinear model predictive control (NMPC) and nonlinear moving horizon estimation (MHE) formulations for controlling the crystal size and shape distribution in a batch crystallization process. MHE is used to estimate unknown states and parameters prior to solving the NMPC problem. Combining these two formulations for a batch process, we obtain an expanding horizon estimation problem and a shrinking horizon model predictive control problem. The batch process has been modeled as a system of differential algebraic equations (DAEs) derived using the population balance model (PBM) and the method of moments. Therefore, the MHE and NMPC formulations lead to DAE-constrained optimization problems that are solved by discretizing the system using Radau collocation on finite elements and optimizing the resulting algebraic nonlinear problem using IPOPT. The performance of the NMPC–MHE approach is analyzed in terms of setpoint change, system noise, and model/plant mismatch, and it is shown to provide better setpoint tracking than an open-loop optimal control strategy. Furthermore, the combined solution time for the MHE and the NMPC formulations is well within the sampling interval, allowing for real world application of the control strategy.

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1. Introduction

Batch crystallization is a crucial process in the pharmaceutical industry because more than 90% of the active pharmaceutical ingredients (API) are in the form of crystals (Alvarez & Myerson, 2010). The crystal size and shape distribution is of great concern to both product quality and downstream processing such as filtration. Therefore, the goal of batch crystallization is to control the crystal qualities to achieve the desired size and shape distribution at the end of the batch process. Primarily because of the technology limitations to monitor the crystal shape (Nagy, Fevotte, Kramer, & Simon, 2013), early works in the crystallization research community focused on modeling and controlling the size distribution of crystals (Mesbah, Kramer, Huesman, & Van den Hof, 2009; Qamar, Mukhtar, Seidel-Morgenstern, & Elsner, 2009). Focused Beam Reflectance Measurements (FBRM) is frequently used to monitor the size distribution online (Braatz, 2002; Fujiwara, Nagy, Chew, & Braatz, 2005; Puel, Févotte, & Klein, 2003). The last decade has witnessed a significant progress in monitoring and modeling the shape distribution of crystals allowing the standard feedback control (Mesbah, Huesman, Kramer, Nagy, & Van den Hof, 2011; Mesbah, Nagy, Huesman, Kramer, & Van den Hof, 2012; Nagy & Braatz, 2003; Patience & Rawlings, 2001; Wan, Wang, & Ma, 2009; Wang, De Anda, & Roberts, 2007). Derived using the multidimensional population balance model (PBM) (Hulburt & Katz, 1964; Ramkrishna,

2000) and the method of moments, the dynamic evolution of the crystal size and shape distribution can be modeled as a system of differential algebraic equations. The size and shape distribution can be controlled by manipulating the cooling profile of the reactor, which directly affects the supersaturation.

To balance the trade-off between the size and shape distribution, Acevedo, Tandy, and Nagy (2015) proposes a multi-objective optimization approach to control both the size and shape distribution offline. However, in the presence of model/plant mismatch and system noise, the real plant trajectory can be quite different from the optimal trajectory obtained from the open-loop multi-objective optimization. Therefore, in this paper, we developed a nonlinear model predictive control (NMPC) formulation that can be used to control the crystal size and shape distribution in real-time and in the presence of modeling and measurement noise.

Linear MPC has been a popular advanced control strategy in industry for many years (Qin & Badgwell, 2003). Because of the advances in both computational power and optimization algorithms, nonlinear model predictive control (NMPC) has become more computationally feasible and is more appropriate for inherently nonlinear systems to achieve higher product quality and satisfy tighter regulations (Mayne, Rawlings, Rao, & Scokaert, 2000; Rawlings, 2000). The basic idea of NMPC is to solve an optimal control problem at each sampling instance with

* Corresponding author.

E-mail address: lairdc@purdue.edu (C.D. Laird).

the updated measured or estimated states. The control values for only the next sampling instance are implemented and the entire process is repeated in the next sampling cycle. For batch processes, since our real interest is in the product quality at the end of the batch, end-point based shrinking horizon NMPC formulation is frequently used.

Nevertheless, for many processes, it is not possible (or cost effective) to accurately measure all states online, and model parameters may change from batch to batch. This challenge drives the need for a state estimator to reconstruct unknown states and parameters. The Extended Kalman filter (EKF) is a popular state estimator for unconstrained systems (Prasad, Schley, Russo, & Bequette, 2002). However, this technique is not appropriate for the batch crystallization model because of the highly nonlinear dynamics and hard constraints such as nonnegative concentrations. In contrast, nonlinear moving horizon estimation (MHE) uses nonlinear constrained optimization to estimate unknown states and parameters and has proven its advantages over EKF in many applications (Haseltine & Rawlings, 2005; Rao, Rawlings, & Mayne, 2003; Rawlings & Bakshi, 2006). Therefore, in this work, we propose an MHE formulation that can be used to estimate the unmeasured states in our model prior to solving the NMPC problem for the batch crystallization process.

The computational burden of this approach is that at each sampling instance, an expanding horizon estimation problem and a shrinking horizon model predictive control problem need to be solved. Both problems are DAE-constrained optimization problems and there exist multiple solution approaches. “Optimize then discretize” or indirect approaches try to solve the first-order optimality conditions for the DAE-constrained problem. For problems without inequality constraints, the first-order optimality conditions can be formulated as boundary value DAE problems. However, for problems with active inequality constraints, determining the switch points of the inequality constraints can become very challenging and thus limits the application of these methods. On the contrary, “discretize then optimize” or direct approaches discretize the control variables and solve the resulting nonlinear programming (NLP) problems. Among “discretize then optimize” approaches, the sequential approach discretizes only the control variables and treats the DAE system as a black box. A DAE integrator is used to simulate the system at each iteration and calculate its sensitivity with regards to the discretized control variables. One drawback of this approach is that the solution time increases significantly when the controls are discretized finer. However, a finer discretization of the controls can often improve the performance of the NMPC. In contrast, the simultaneous approach (Biegler, 2007; Biegler, Cervantes, & Wächter, 2002) discretizes both control and state variables and optimizes the resulting algebraic nonlinear problem with an NLP solver. The performance of the simultaneous approach is less dependent on the number of discretized control variables. Another advantage of this approach is that state constraints can be formulated in a more straightforward way. Therefore, this paper chooses the simultaneous approach to solve these DAE-constrained optimization problems arising from the NMPC–MHE formulations. One challenge of using the simultaneous approach is that the burden of manually discretizing the DAE system before it is embedded into an optimization formulation often lies on the user. However, packages such as the Modelica-based JModelica.org platform (Åkesson, Årzén, Gäfvert, Bergdahl, & Tummescheit, 2010) allow for straightforward declaration of differential equations and automatically perform this transcription process. Therefore, we implement these control formulations for batch crystallization within the Modelica library, which is already interfaced with solvers like IPOPT. If IPOPT is not fast enough, some parallel solvers can potential be used (Cao, Laird, & Zavala, 2016; Cao, Seth, & Laird, 2016; Kang, Cao, Word, & Laird, 2014).

This paper is organized as follows: a description of the unseeded batch crystallization model is presented in Section 2. Section 3 presents the NMPC–MHE approaches and efficient methods to solve the related optimization problems. Section 4 demonstrates the performance of the NMPC–MHE compared with the open-loop control in terms of setpoint change, system noise, and model/plant mismatch. Final conclusions are presented in Section 5.

2. Multidimensional unseeded batch crystallization model

This section provides a brief description of the multidimensional unseeded batch crystallization model. The details can be found in Acevedo and Nagy (2014). The population balance model (PBM) has been widely used to describe the crystallization process (Cao, Kang, Nagy, & Laird, 2016; Mesbah et al., 2012). Considering only the effect of growth and nucleation, the population balance equation for a well-mixed batch crystallization process can be expressed as

$$\frac{\partial}{\partial t} n(t, X) + \nabla_X [Gn(t, X)] = B\delta(X - X_0) \quad (1a)$$

$$n(0, X) = n_0(X), \quad (1b)$$

where $n(t, X)$ is the density distribution at time t , X is the vector of characteristic lengths, G is the vector of growth rates, B is the nucleation rate, X_0 is the size of the nuclei, δ is the Dirac delta function acting at $X = X_0$, and $n_0(X)$ is the initial seed distribution. The population balance model can be transformed into a set of ordinary differential equations (ODEs) using the method of moments (MOM). If we only consider two characteristic dimensions, the length L and the width W of crystals, the moments can be expressed by

$$\mu_{ij} = \int_0^\infty \int_0^\infty n(t, X) W^i L^j dW dL. \quad (2a)$$

The ODEs obtained from the MOM with the assumption that the nucleus size is negligible, are given by

$$\frac{d\mu_{00}}{dt} = B \quad (3a)$$

$$\frac{d\mu_{10}}{dt} = G_1 \mu_{00} \quad (3b)$$

$$\frac{d\mu_{01}}{dt} = G_2 \mu_{00} \quad (3c)$$

$$\frac{d\mu_{11}}{dt} = G_1 \mu_{01} + G_2 \mu_{10} \quad (3d)$$

$$\frac{d\mu_{20}}{dt} = 2G_1 \mu_{10}, \quad (3e)$$

where G_1 and G_2 are the growth rates along the width and length of the crystals respectively, and B is the nucleation rate. In this work, size independent growth rates and primary nucleation rate are considered as follows:

$$G_1 = k_{g1} S^{g_1} \quad (4a)$$

$$G_2 = k_{g2} S^{g_2} \quad (4b)$$

$$B = k_b S^b \quad (4c)$$

$$S = \frac{C - C_s(T)}{C_s(T)}, \quad (4d)$$

where the kinetic parameters k_{g1} , k_{g2} , g_1 , g_2 , k_b , and b are usually sensitive to process conditions. S is the relative supersaturation, C is the solute concentration, and C_s is the equilibrium concentration at a given temperature, which can be expressed using a polynomial expression, given by

$$C_s(T) = cT^2 + dT + e. \quad (5a)$$

According to the mass balance equation, the evolution of the solute concentrate is given by

$$\frac{dC}{dt} = -2\rho_c k_v G_1 (\mu_{11} - \mu_{20}) - \rho_c k_v G_2 \mu_{20}, \quad (6a)$$

where ρ_c is the density of the solution and k_v is a constant volumetric shape factor.

3. Computationally efficient online NMPC–MHE

At the end of the batch crystallization process, the product qualities are evaluated in terms of the size and shape distribution of crystals. Therefore, the mean length (ML) and aspect ratio (AR) are used to

evaluate the quality of crystals. ML can be calculated with the following equation

$$ML = \frac{\mu_{01}}{\mu_{00}}, \quad (7a)$$

and AR is determined by the following equation

$$AR = \frac{\mu_{01}}{\mu_{10}}. \quad (8a)$$

The product qualities are determined by the supersaturation trajectory, which is dependent on the temperature profile. Thus, the temperature profile can be used to control the crystal qualities to achieve the desired size and shape distribution.

3.1. Off-line multi-objective optimization

Before implementing the control strategy, we need to set an endpoint target. To find achievable end-point setpoints for the ideal case, we first solve the multi-objective optimization problems off-line. The problems follow the formulations in Acevedo et al. (2015) and Ma, Tafti, and Braatz (2002),

$$\min_{u(t)} \varphi(y(t_f)) \quad (9a)$$

$$\text{s.t.} \quad \frac{dx(t)}{dt} = f(x(t), u(t)) \quad (9b)$$

$$y(t) = c(x(t), u(t)) \quad (9c)$$

$$x(t_0) = \tilde{x}_0 \quad (9d)$$

$$g(x(t), u(t)) \leq 0, t \in [t_0, t_f], \quad (9e)$$

where t is the time, t_0 and t_f are the start time and end time of the process, $x(t)$ is the vector of state variables including differential variables and algebraic variables, $y(t)$ is the vector of output variables AR , ML and C , and u represents the manipulated variable. Specifically for the crystallization process, x includes $\mu_{00}, \mu_{10}, \mu_{01}, \mu_{11}, \mu_{20}$, $y(t)$ represents AR , ML and C , and u is the temperature T . $c(x(t), u(t))$ is the system model including all the equations introduced in Section 2.

The initial state values $\tilde{x}_0$ of the process is known. Eq. (9e) is a vector of constraints on the inputs and state variables that can be further detailed using the following set of equations:

$$T_{min} \leq T(t) \leq T_{max} \quad (10a)$$

$$-R_{max} \leq \frac{dT(t)}{dt} \leq 0, \quad (10b)$$

$$C(t_f) - C_{max} \leq 0. \quad (10c)$$

Eqs. (10a) and (10b) ensure that changes of temperature are within the operation range. Eq. (10c) is the yield constraint.

For batch crystallization processes, we want to avoid needle crystals. Therefore, we want AR to be small and ML to be large. The objective function $\varphi(y(t_f))$ is defined as

$$\varphi(y(t_f)) = (1 - w)AR - wML. \quad (11)$$

With a set of weight values $0 < w < 1$, we can calculate a set of non-dominated points. Without the existence of any model/measurement noise or model/plant mismatch, each Pareto point is achievable using either the open-loop control or NMPC. Therefore, we should choose points on or above the Pareto front as the setpoint.

3.2. Endpoint-based shrinking horizon NMPC formulation

NMPC uses a nonlinear model to predict the dynamic behavior of the system and thus to determine the optimal input profile. This section describes the endpoint based shrinking horizon NMPC used in this paper. The whole process interval $[t_0, t_f]$ can be discretized into N

steps. At a sampling instance t_k , the following optimal control problem is solved online:

$$\min_{u(t)} \|y(t_f) - y_s\|_Q^2 \quad (12a)$$

$$\text{s.t.} \quad \frac{dx(t)}{dt} = f(x(t), u(t)) \quad (12b)$$

$$y(t) = c(x(t), u(t)) \quad (12c)$$

$$x(t_k) = \hat{x}(t_k) \quad (12d)$$

$$g(x(t), u(t)) \leq 0, t \in [t_k, t_f], \quad (12e)$$

where $\hat{x}(t_k)$ is a vector of states at t_k estimated from measurements using MHE and y_s is the setpoint we want to achieve at the end of the batch based on the off-line multi-objective optimization. Although the whole input profile in the interval $[t_k, t_f]$ is computed, only the control action in the interval $[t_k, t_{k+1}]$ is implemented. At the next sampling instance t_{k+1} , the control horizon shrinks from $[t_k, t_f]$ to $[t_{k+1}, t_f]$, and the optimal control problem is re-evaluated with new measurements and updated state estimates.

For end-point based NMPC, the objective function only depends on $x(t_f)$. Here, we want to minimize the deviation of the product quality at the end of the batch from the desired product qualities. The deviation is weighted by a positive definite matrix Q .

3.3. Nonlinear expanding horizon MHE formulation

Since the NMPC approach requires that the initial values of all state variables $\hat{x}(t_k)$ are known, and since not all states can be measured, we need to reconstruct the state information from a limited number of measurements. At each sampling instance t_k , a state estimation problem is solved online and the solution $\hat{x}(t_k)$ is provided to the NMPC formulation. The objective function of the optimization problem penalizes the model noise and the deviation of predicted output from the measurements and the estimated parameters from the reference value. The formulation of the estimation problem is as follows:

$$\min_{p, w(t)} \int_{t_0}^{t_k} \|w(t)\|_R^2 dt + \sum_{i=1}^k \|y(t_i) - y(t_i)^{meas}\|_W^2 + \|p - p^{ref}\|_Z^2 \quad (13a)$$

$$\text{s.t.} \quad \frac{dx(t)}{dt} = f(x(t), u(t), p) + w(t) \quad (13b)$$

$$y(t) = c(x(t), u(t)) \quad (13c)$$

$$x(t_0) = \tilde{x}_0 \quad (13d)$$

$$x(t) \geq 0, t \in [t_0, t_k], \quad (13e)$$

where $y(t_i)^{meas}$ is the set of actual measured values at sampling instance t_i , w is the vector of model noise, p is the vector of parameters selected for online adjustment, p^{ref} is the vector of reference value for p , and R , W and Z are weighting matrices. Generally, the objective function of MHE also includes a term about the initial state estimation. However, for this unseeded batch process, the initial state values are known. The time span of this problem is $[t_0, t_k]$, therefore the entire historical input profile is known. At the next sampling instance t_{k+1} , the time span of the problem expands from $[t_0, t_k]$ to $[t_0, t_{k+1}]$, and the estimation problem is re-evaluated with new measurements $y(t_{k+1})^{meas}$.

3.4. Efficient optimization via the simultaneous approach

The off-line multi-objective optimization problem, the NMPC problem and MHE problem are all DAE-constrained optimization problems. These problems can be solved using the simultaneous method by discretizing the problem with collocation methods and optimizing the large NLP problem afterwards. The details of this method can be found in Biegler (2010). This section uses the NMPC problem as an example to give a brief description of this method. This approach partitions the time domain $[t_k, t_f]$ into n_c stages with length h_i , $i = 1, \dots, n_c$, where $\sum_{i=1}^{n_c} h_i = t_f - t_k$. At each stage, we discretize using n_c collocation points.

Table 1
Parameters used in the control of unseeded cooling batch crystallization systems

Parameters	Values	Units	Parameters	Values	Units
k_{g1}	0.073	cm/min	T_{max}	45	°C
g_1	1.48	Dimensionless	T_{min}	5	°C
k_{g2}	0.60	cm/min	R_{max}	−4	°C/min
g_2	1.74	Dimensionless	C_0	0.395	g/cm ³
k_b	$4.494 \cdot 10^6$	#/cm ³ min	C_{max}	0.2	g/cm ³
b	2.04	Dimensionless	t_f	90	min
k_v	0.67	Dimensionless	ρ_c	2.34	g/cm ³

This section assumes Radau collocation is used. After discretization, the problem can be formulated as:

$$\min_{u_{i,j}, x_{i,j}, y_{i,j}, \dot{x}_{i,j}} \|y_{n_c, n_c} - y_s\|_Q^2 \quad (14a)$$

$$\text{s.t. } x_{i,j} = x_i + h_i \sum_{k=1}^{n_c} w_{j,k} \dot{x}_{i,j} \quad (14b)$$

$$\dot{x}_{i,j} = f(x_{i,j}, u_{i,j}) \quad (14c)$$

$$y_{i,j} = c(x_{i,j}, u_{i,j}) \quad (14d)$$

$$x_1 = \hat{x}(t_k) \quad (14e)$$

$$x_{i+1} = x_{i, n_c} \quad (14f)$$

$$g(x_{i,j}, u_{i,j}) \leq 0 \quad (14g)$$

$$\forall i = 1, \dots, n_c, j = 1, \dots, n_c, \quad (14h)$$

where w are the coefficients from the Radau collocation method.

One challenge of using the simultaneous approach is manual discretization of the problem. However, the Modelica-based JModelica.org platform (Åkesson et al., 2010) allows for straightforward declaration of differential equations and automatically perform this transcription process using direct collocation methods. Therefore, we implement the control formulation for batch crystallization within the Modelica library, which is already interfaced with solvers like IPOPT.

4. Results and discussion

In the results shown later, the potassium dihydrogen phosphate (KH_2PO_4 , KDP) and water system is used as a case study. The equilibrium concentration C_s for KDP estimated by Togkalidou, Fujiwara, Patel, and Braatz (2000, 2001) is given by

$$C_s(T) = 9.3027 \times 10^{-5} T^2 - 9.7629 \times 10^{-5} T + 0.2087, \quad (15)$$

where the unit of T is °C and the unit of C_s is g/cm³. The kinetic parameters of KDP (Acevedo & Nagy, 2014; Gunawan, Ma, Fujiwara, & Braatz, 2002; Majumder & Nagy, 2013) and process conditions are summarized in Table 1.

Before implementing the control strategies, we first find achievable end-point setpoints by solving open-loop multi-objective optimization problems offline. Two cases are considered with the control variable T discretized with 6 control steps and 90 control steps. The temperature profile inside the same step is assumed to be linear. The state variables are discretized with 90 steps. For the larger case (using 90 control steps), the number of variables and constraints in each problem are 5320 and 5230 respectively. All the problems are initialized using the simulation data with linear temperature profile. We use a computing server with Intel(R) Xeon(R) CPU E5-2698 v3 processors running at 2.30 GHz. The solution time for IPOPT on an individual Pareto point is approximately 2 s. By solving the problem repeatedly with different weights we obtain the Pareto front. Weights are generated using equidistant grid. Fig. 1 clearly demonstrate the trade-off between the two objectives. We can also generate this figure using epsilon constraint method. The front obtained using 6 control steps is worse than that using 90 control steps since it has fewer degrees of freedom.

All the points below the Pareto front are not achievable in the ideal circumstance with no system noise or model/plant mismatch. Therefore

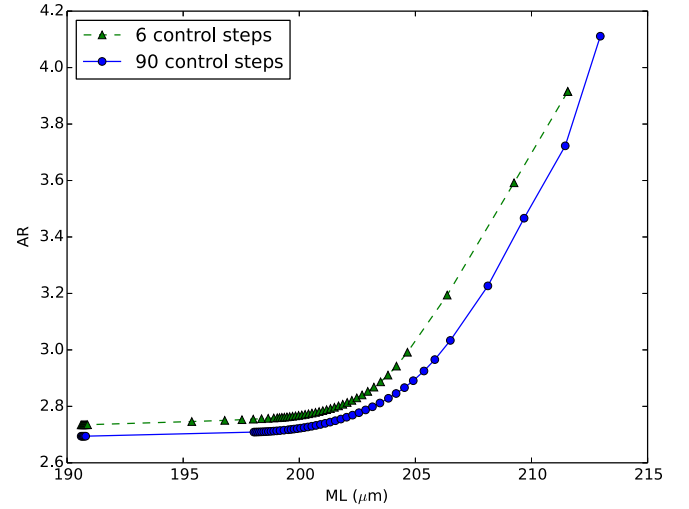


Fig. 1. Pareto fronts between AR and ML using 6 and 90 control steps.

Table 2
Effect of t_{change} and sampling/control steps on end-point performance.

Sampling/Control steps	t_{change} (min)	AR	ML (μm)	cost
6	0	3.880	210.65	0.00
	30	3.358	198.10	185.8
	60	2.884	195.32	335.7
	90	2.735	190.53	537.8
18	0	3.883	210.66	0.0
	30	3.533	197.67	181.5
	60	2.784	197.26	300.8
	90	2.735	190.53	537.8
90	0	3.882	210.65	0.0
	30	3.439	197.97	181.2
	60	2.769	197.47	298.6
	90	2.735	190.53	537.8

we should choose a point above the Pareto front as the setpoint, which is then used to construct the objective function in the NMPC. The objective function used in the NMPC for this application is

$$cost = 100(AR(t_f) - AR_s)^2 + (ML(t_f) - ML_s)^2, \quad (16)$$

where AR_s and ML_s are the end-point setpoints. Since the scale of AR_s and ML_s , a weight of AR is introduced in the cost function. This cost function is also used to judge the performance of different control approaches. We analyze the performance of the NMPC–MHE during setpoint change, system noise and model/plant mismatch.

4.1. Setpoint change

Although setpoint change during the batch process is uncommon, we use this study to examine the performance of our closed-loop NMPC–MHE. We first select two points on the Pareto front with 6 control steps as setpoints so that it is more fair to compare the performance with different control steps. Here we choose $AR_{s1} = 2.735$, $ML_{s1} = 190.53$ μm for setpoint s_1 and $AR_{s2} = 3.883$, $ML_{s2} = 210.68$ μm for setpoint s_2 . At a certain time during the process, the setpoint is changed from s_1 to s_2 . For the results discussed in this subsection, we assume all states are exactly measured and there is no system noise and model/plant mismatch.

Fig. 2 shows the input and measurement profiles when the setpoint is changed at $t = 30$ min. The number of control and sampling steps are all 90. Before the setpoint change, the NMPC profile follows the open-loop trajectory for achieving s_1 . However, after the setpoint change, the NMPC profile moves closer to the open-loop trajectory for achieving s_2 .

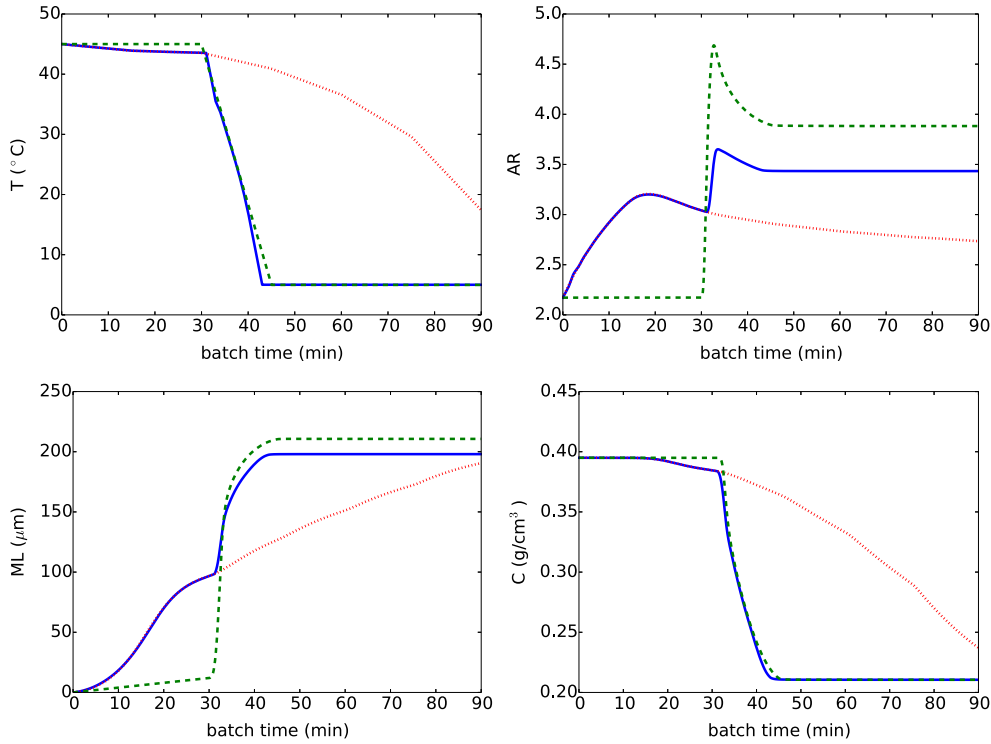


Fig. 2. Input and measurement profiles when the setpoint is changed at $t = 30$ min. The solid line denotes the NMPC profile, the dotted line denotes the open-loop trajectory achieving endpoint setpoint s_1 , and the dashed line denotes the open-loop trajectory achieving endpoint setpoint s_2 . Before $t = 30$ min, the NMPC profile follows the dotted line, while after setpoint change, the NMPC profile moves closer to the dashed line.

Table 2 shows the effect of t_{change} and sampling/control steps, where t_{change} is the time setpoint changed. The endpoint product quality of the NMPC is closer to setpoint s_2 when the setpoint change is performed earlier in the process. It also indicates that increasing the number of sampling/control intervals can improve the performance of NMPC slightly in this case.

4.2. System noise

This subsection demonstrates the effectiveness of the closed-loop NMPC–MHE for the batch process with both model and measurement noise. We assume that there is one model noise term w added to $\frac{d\mu_{01}(t)}{dt}$ and the noise follows the truncated normal distribution on the interval $[-20 \ 20]$ cm/cm³ min. The mean and standard deviation of the original normal distribution is 0 and 10 cm/cm³ min respectively. We also assume that there is online measurements of ML , AR and C and the measurement noise corresponding to ML , AR and C follow the truncated normal distribution on the interval $[-6 \ 6]$ μ m, $[-0.1 \ 0.1]$, and $[-0.004 \ 0.004]$ g/cm³. The mean values of the original normal distribution are all zero, and the standard deviations are 3 μ m, 0.05, and 0.002 g/cm³, respectively. Because of the noise, points on the Pareto front can no longer be achieved. Therefore, we consider a more conservative setpoint s_3 , where AR_{s_3} is 2.9 and ML_{s_3} is 195 μ m.

Fig. 3 shows the relative estimation error of states using MHE with 90 control and sampling steps for one noise scenario. This figure shows that MHE can reconstruct the evolution of the state for this batch process fairly accurately. The relative estimation errors stabilize after about 15 min with enough data.

Table 3 highlights the performance improvement that can be achieved from the NMPC–MHE approach. It shows that the performance of ideal NMPC with all states accurately measured is much better than that of the open-loop control. The performance of NMPC–MHE is very close to that of the NMPC with true states and is also much better than that of the open-loop control. This observation is only possible with the excellent performance of MHE to reconstruct unmeasured states and

thus give accurate feedback. This table also indicates that increasing the number of control and sampling steps can greatly improve the performance of NMPC–MHE.

Fig. 4 shows that the CPU computational time of the expanding horizon estimation problems increases along the batch process, and that of the shrinking horizon model predictive control problems decreases. Because of the efficient computational framework used, the maximum total computation time (approximately 7 s) is far below the sampling interval of 60 s, allowing for real world application of the proposed control strategy.

4.3. Model/plant mismatch

This section considers the case with both model/plant mismatch and measurement noise. It is assumed that the actual value in the plant for the parameter kb is $5.494 \cdot 10^6$, while the initial guess used in the first NMPC instance is $4.494 \cdot 10^6$. By using the MHE, we not only reconstruct the state profiles from measurements, but also estimate the unknown parameter. The unknown parameter and measurement noise become degrees of freedom in the MHE. Another term $\|kb - 4.494 \cdot 10^6\|^2$ is added to the objective function for regularization inside the MHE. The measurement noise and setpoint are the same as that in Section 4.2.

The estimated parameter and state profiles from MHE are used in the NMPC at the same sampling instance. Therefore, the accuracy of the state profile and parameter estimation is very important to the performance of NMPC–MHE. The state profile estimation results are still as accurate as the results shown in Section 4.2. However, the parameter estimation, as shown in Fig. 5, is not very accurate over the first 15 min but gradually converges to the true value as more measurement data becomes available.

Table 4 shows the overall performance of NMPC–MHE in dealing with model/plant mismatch. Again, the performance of NMPC–MHE is much better than that of open-loop control. Increasing the number of control and sampling steps can improve the performance of NMPC–MHE in the presence of model/plant mismatch. This table also considers the

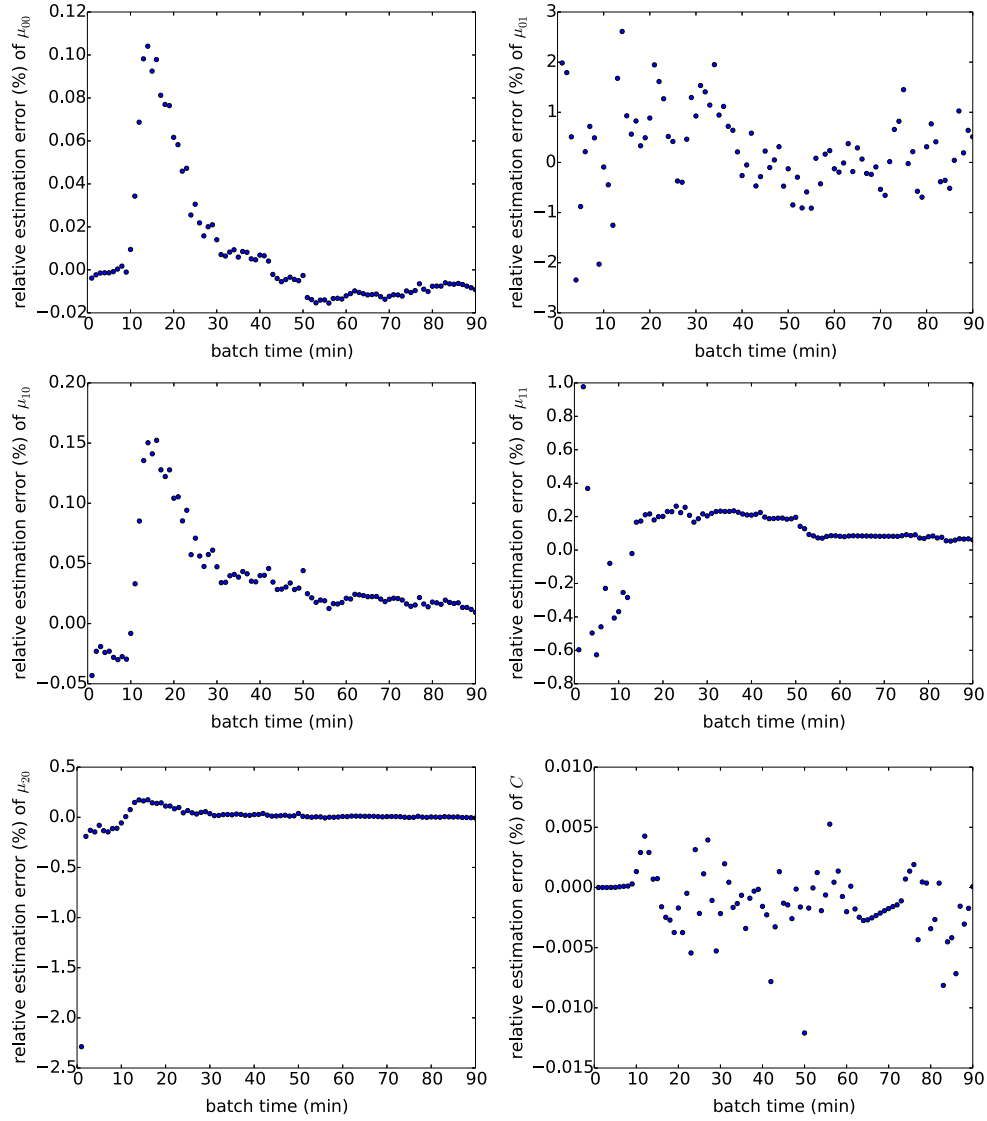


Fig. 3. Evolution of the relative estimation error of states using MHE with 90 control and sampling steps.

Table 3

Performance of NMPC–MHE (value of *cost*) on 10 cases with model and measurement noise. Closed loop with true states is the performance of NMPC with 90 control and sampling steps and all states exactly measured.

Scenario no.	Open loop (<i>cost</i>)	Closed loop (<i>cost</i>)			
	90 steps	6 steps	18 steps	90 steps	True states
1	159	31.4	24.8	18.2	17.7
2	14.5	1.0	0.9	1.0	1.1
3	105.5	33.1	40.7	40.6	42.9
4	52.4	27.7	17.0	12.0	12.7
5	297.2	200.1	154.4	140.4	116.9
6	53.0	22.5	17.8	9.7	10.5
7	28.4	1.4	3.0	3.8	2.7
8	44.5	16.4	8.5	3.9	2.7
9	81.4	12.6	10.7	9.3	9.4
10	13.7	24.8	8.9	1.1	3.2
Average	85.0	37.1	28.7	23.9	22.0

performance of NMPC with all states accurately measured and a fixed initial guess (inaccurate) of the parameter *kb*. Comparing with NMPC–MHE, this NMPC has exact state measurement but has no parameter estimation updates. The performance of NMPC–MHE is slightly better than that of the NMPC with true states, indicating the importance of parameter updates.

5. Conclusions and future work

In summary, we have developed computationally effective NMPC–MHE formulations for batch crystallization processes to control the crystal size and shape distribution. At each sampling instance, we need to solve an expanding horizon estimation problem and a shrinking horizon

Table 4

Performance of NMPC–MHE (value of *cost*) on 10 cases with model/plant mismatch and measurement noise. Closed loop with 6 steps, 18 steps, and 90 steps are the performance of NMPC with state estimation and parameter updates from MHE. Closed loop with true states is the performance of NMPC with 90 control and sampling steps and all states exactly measured. However, the parameter *kb* is fixed to be $4.494 \cdot 10^6$, which is not accurate.

Scenario no.	Open loop (<i>cost</i>)	Closed loop (<i>cost</i>)			
	90 steps	6 steps	18 steps	90 steps	True States
1	73.5	7.7	7.6	3.4	6.8
2	73.5	5.3	7.5	4.5	6.8
3	73.5	7.4	7.5	2.9	6.8
4	73.5	5.2	4.0	4.0	6.8
5	73.5	8.6	9.7	3.5	6.8
6	73.5	8.6	5.0	3.7	6.8
7	73.5	7.3	5.6	3.3	6.8
8	73.5	6.4	7.6	3.8	6.8
9	73.5	7.7	7.7	4.9	6.8
10	73.5	7.3	7.6	5.9	6.8
Average	73.5	7.2	7.0	4.0	6.8

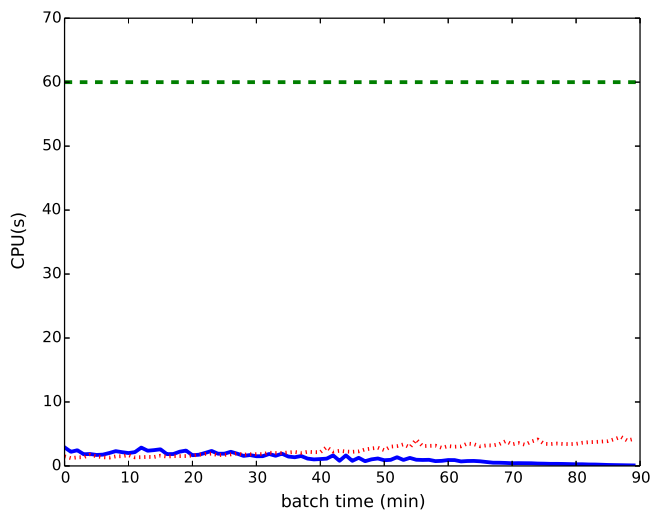


Fig. 4. Computational time of NMPC (solid line), computational time of MHE (dotted line), and sampling interval (dash line) along the batch with 90 control and sampling steps.

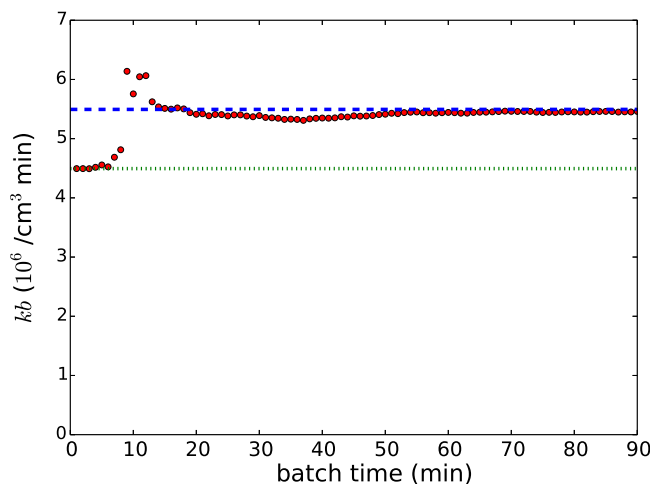


Fig. 5. Actual value (dash line), initial guess (dotted line), and MHE estimation (dots) of parameter *kb* along the batch process with 90 control and sampling steps.

model predictive control problem. Based on a nonlinear DAE model, the estimation problem estimates unknown states and parameters and the control problem determines the optimal input profiles. Both DAE-constrained optimization problems are solved by discretizing the system

using Radau collocation and optimizing the resulting algebraic nonlinear problem. We build these formulations in the Modelica modeling language to support solution through the JModelica modeling and optimization framework. This framework performs automatic transcription, and it is already interfaced with IPOPT.

The performance of this control strategy was tested using a case study of a 90-min batch crystallization process with 90 control steps and sampling steps. It was analyzed in terms of setpoint change, system noise, and model/plant mismatch. For all cases, the performance of the NMPC–MHE is shown to provide better setpoint tracking than the open-loop optimal control strategy. The combined solution time for the MHE and the NMPC formulations is well within the sampling interval, allowing for real world application of the control strategy.

In the future, we would extend this work to alternative formulations, e.g. the formulations with nucleation in the boundary conditions. The model we used is based on assumptions like size independent growth rate. We would further extend the NMPC–MHE formulation to more complicated cases using quadrature method of moments and compare with other closed-loop control approaches.

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